



Non-blocking Sockets



About "fcntl()": <https://www.man7.org/linux/man-pages/man2/fcntl.2.html>

- Normal mode:
 - `recv()` waits forever until data arrives
 - This blocks the Web Server
- Non-blocking mode:
 - `recv()` returns immediately with -1 and "errno = EAGAIN" if no data.

```
#include <fcntl.h>

int flags = fcntl(socketFd, F_GETFL, 0);
fcntl(socketFd, F_SETFL, flags | O_NONBLOCK);
```

On MacOS, "write()" behaves differently. In which case "fcntl()" can be called with:

- F_SETFL
- O_NONBLOCK

- FD_CLOEXEC (close on exec, for CGI safety)

```
// MacOS allowed flags only
fcntl(fd, F_SETFL, O_NONBLOCK);
fcntl(fd, F_SETFD, FD_CLOEXEC);
```